

Education

- 2015-2020 **Dual Degree (B.Tech + M.Tech) in Computer Science and Engineering**
Indian Institute of Technology Madras, Chennai, India CGPA: 8.78
- 2015 **XII - Karnataka Board, KLE Society's Independent PU College, Bangalore** 97.30 %
- 2013 **X - ICSE, B P Indian Public School, Bangalore** 96.33%

Professional Experience

- Software Engineer, Level IV, Google LLC, New York**
- Dec 2022 - Present **Software Engineer, Level IV, Google India Pvt Ltd, Bangalore**
- Aug 2020 - Nov 2022 { Developing intelligent features for the Google Workspace Editors (Docs, Slides, Keep, etc) using my expertise on the products' client-side software, supporting tools and libraries, and natural language processing infrastructure.
 { Using cutting-edge frontend tools like Web Assembly and Emscripten, and Google-internal technologies like j2Cl, client-side cross-platform frameworks and build systems, to develop user-facing features such as spellcheck in encrypted documents for five languages and writing style suggestions for English text.
 { Formulating technical designs for independent end-to-end problems, driving cross-team collaboration, upholding software reliability practices, technical-debt resolution and documentation, and proactively identifying areas of future work.
 { Guiding junior engineers on programming and software design tasks to enable timely delivery of products to customers.
- May 2019 - July 2019 **Software Engineering Intern, Google India Pvt Ltd, Bangalore**
- Worked on the Editors client-side software infrastructure to develop a user interface with control options to undo or provide feedback on the correction and a logging framework, for the Google Docs text auto-correction feature.
- May 2018 - July 2018 **Research Intern, Big Data Experience Labs, Adobe Research, Bangalore**
- Developed a mobile application for Text to Scene Conversion in Augmented Reality, based on novel research techniques for prediction of three-dimensional object sizes and positions from textual features.

Research Experience

- Sep 2019 - May 2020 **Paraphrase Generation with a Bilingual Model and Continuous Embeddings** *Master's Thesis, Language Technologies Institute, Carnegie Mellon University*
- Machinated a novel technique for paraphrase generation using the [von Mises-Fisher \(vMF\) Loss](#) on a transformer network, and showed that it produces superior paraphrases as compared to the log-likelihood model by employing bilingual data to induce zero-shot paraphrasing, guided by [Prof. Yulia Tsvetkov](#).
- May 2017 - July 2017 **Cognitive Approach to Natural Language Processing** *Research Intern, Department of Computer Science and Automation, Indian Institute of Science (IISc), Bangalore*
- Developed a cognitive text parser that combines syntactic and semantic approaches, to process textual data into cognitive structural representations, to be used as a feature extractor for downstream NLP tasks, and demonstrated the correlation of the extracted cognitive features with semantic and syntactic text features, guided by [Prof. Veni Madhavan](#).

Publications and Patents

- [Publication and Poster] **Improving the Diversity of Unsupervised Paraphrasing with Embedding Outputs (Paper, Poster)**
Monisha Jegadeesan, Sachin Kumar, John Wieting, Yulia Tsvetkov
 In [Workshop on Multilingual Representation Learning](#),
 The 2021 Conference on Empirical Methods in Natural Language Processing ([EMNLP 2021](#))
- [Publication and Poster] **Adversarial Demotion of Gender Bias in Natural Language Generation (Paper, Poster)**
Monisha Jegadeesan
 In [ACM CODS-COMAD 2020](#) - Young Researchers' Symposium
- [Poster] **ARComposer: Authoring Augmented Reality Experiences through Text (Poster)**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 In ACM User Interface Software and Technology Symposium 2019 ([ACM UIST 2019](#))
- [Filed Patent] **Visualizing Natural Language through 3D Scenes in Augmented Reality**
Sumit Kumar, Paridhi Maheshwari, Monisha Jegadeesan, Amrit Singhal, Kush Kumar Singh, Kundan Krishna
 Filed at the US PTO (Application Number: 16/247,235)

Projects

Real-Time AI Chatbot for Customer Support (2024)

Developed an AI-powered chatbot by structuring real-world customer support workflows into clear task specifications and datasets, enhancing the retrieval-augmented generation (RAG) pipeline to improve AI system reasoning. Created high-quality reference examples reflecting professional standards for AI integration, significantly reducing human agent workload by 35% and improving response time from 5 minutes to instant replies. Collaborated asynchronously using structured documentation and delivered a robust solution aligned with consulting problem-structuring and evaluation rubric frameworks.

Tech: LangChain, OpenAI API, ChromaDB, Python

Call Center Performance Analytics Dashboard (2024)

Transformed end-to-end operational workflows into structured data analysis tasks, designing relational SQL queries and KPI evaluation rubrics to benchmark call center performance against strategic objectives. Developed an interactive Power BI dashboard to synthesize agent performance metrics and SLA compliance, enabling hypothesis-driven problem solving and executive-level synthesis that led to a 22% reduction in wait time and improvement in SLA compliance from 78% to 91%. Delivered clear, structured analytical models and slide outlines communicated asynchronously to stakeholders.

Tech: SQL, Power BI, Excel, DAX

Employee Attrition Predictive Analytics (2024)

Employed qualitative research and quantitative modeling to break down HR analytics workflows into defined tasks and evaluation criteria, identifying key attrition drivers such as salary dissatisfaction and overtime through exploratory data analysis and logistic regression models. Created high-quality HR dashboards serving as professional consulting standard references, which informed policy changes that reduced attrition by 12%. Documented logical gaps and improvement guidance to refine AI-assisted analysis efforts under a deliverables-driven, structured thinking approach.

Tech: SQL, Power BI, Python

Website Traffic and Conversion Analytics (2024)

Structured web traffic and user behavior analysis workflows into clear evaluation rubrics and task specifications aligned with consulting problem-structuring methodologies. Developed funnel analysis dashboards and identified bounce rate drivers through layered data segmentation, enabling strategic recommendations to improve conversion rates by 17% and reduce bounce rates by 10%. Delivered detailed analytical reports and slide outlines asynchronously, demonstrating advanced skills in hypothesis-driven problem solving and structured communication.

Tech: Google Analytics, SQL, Power BI

About :

Experienced professional focused on leveraging structured problem-solving frameworks and hypothesis-driven analysis to improve AI systems' reasoning and decision-making capabilities. Skilled at translating complex workflows into clear task specifications, evaluation rubrics, and high-quality reference materials that meet professional consulting standards. Proven expertise in reviewing, benchmarking, and enhancing AI-generated strategic outputs such as market analyses, operating models, and executive summaries. Strong written communication, structured thinking, and remote collaboration abilities, with a deliverables-driven approach and excellent time management skills. Dedicated to integrating AI technologies to automate workflows and support asynchronous teamwork in fully remote environments.

